

Why Pure-Classical IVP?

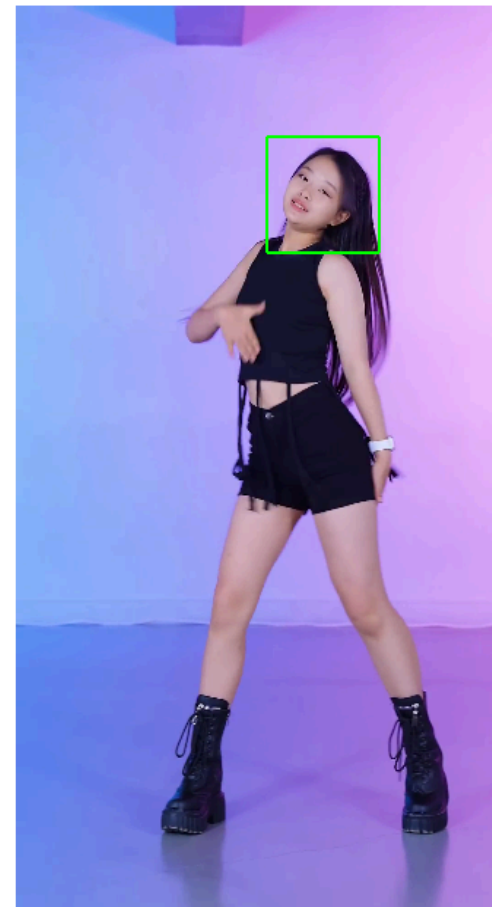
Neural approaches fall short

- **GAN stylization (VToonify)** — polished but *inexplicable*; cannot defend per-pixel
- **Full-face replacement (DeepPrivacy)** — subject loses *agency*, can't recognize self
- **AR filters (Snapchat / TikTok)** — cosmetic, not formal anonymization
- **Black-box models** — heavy lifting hidden, every pixel transformation un-defendable

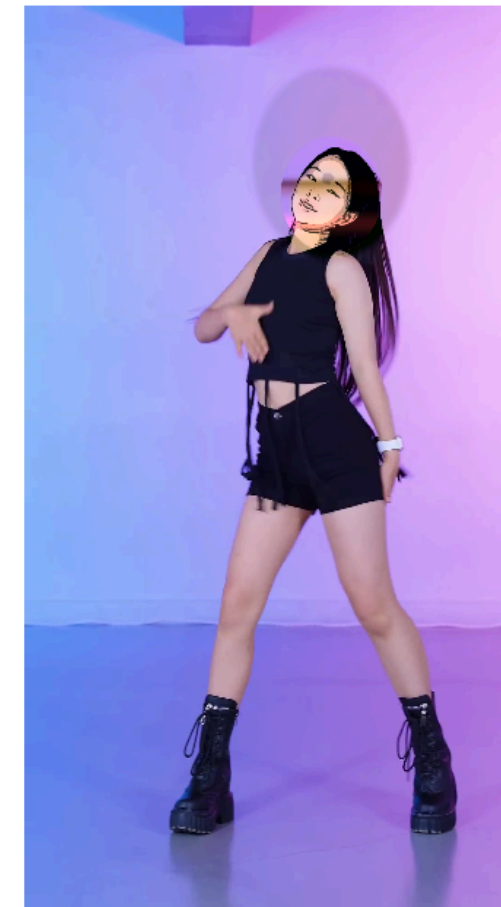
Pure-classical advantages

- **OpenCV + NumPy + Pillow only** — 0 GAN, 0 MediaPipe, 0 neural model
- **CPU-only** ~2.7 fps on M1
- **Fully explainable** — every stage is a textbook IVP operator
- **Identity reborn** from theme palette + self-recognition cues (preserve luminance)

INPUT (Yejin, 13yo)

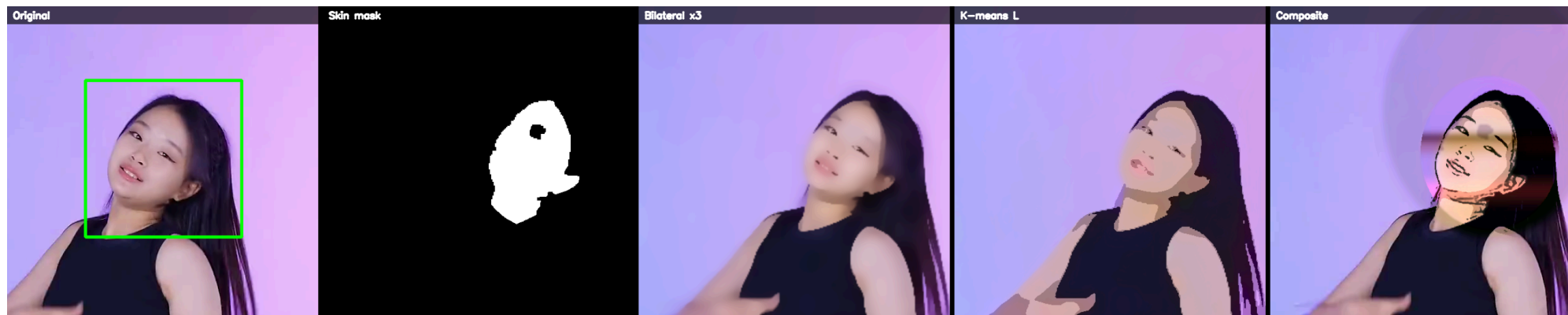


AvatarShield IVP cel-sh



Same frame · cel-shade core + porcelain-pink palette · ~2.7 fps M1 CPU

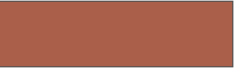
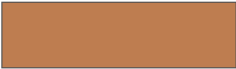
Pipeline Overview



Stage	Module
Detect	Haar cascade (frontal + profile)
Skin mask	$\text{HSV} \cap \text{YCbCr} + \text{morphology} + \text{connected components}$
Cel-shade core	$\text{Bilateral} \times 3 \rightarrow \text{Lab K-means} \rightarrow \text{XDoG} \rightarrow \text{HSV lift}$
Recolor	Reinhard transfer in Lab, per region (hair chroma-only)
Stabilize	One Euro on bbox + EMA $\alpha=0.85$ on K-means centers
Composite	Alpha blend ellipse onto original frame

Every stage is a transparent, well-known IVP operator — no "magic neural step". Defendable end-to-end.

5 Theme Palettes × Per-Region Reinhard

	SKIN	HAIR	LIPS	EYES	BROWS
porcelain-pink					
tan-amber					
ivory-violet					
bronze-teal					
peach-noir					

Palette structure — 5 Lab triples (skin · lips · eyes · brows · hair) + 4 style scalars (luminance levels · edge strength · saturation boost · chroma levels) per theme. Reinhard transfer in **Lab space, per region**.

Hair: chroma-only — preserves user's L → self-recognition cue intact. Subject sees *a stylized themselves*, not a stranger; strangers see a scrambled identity.

Temporal Stability — One Euro + EMA



EMA off (left) vs EMA $\alpha=0.85$ (right) — same source, no boiling

Two anti-flicker mechanisms

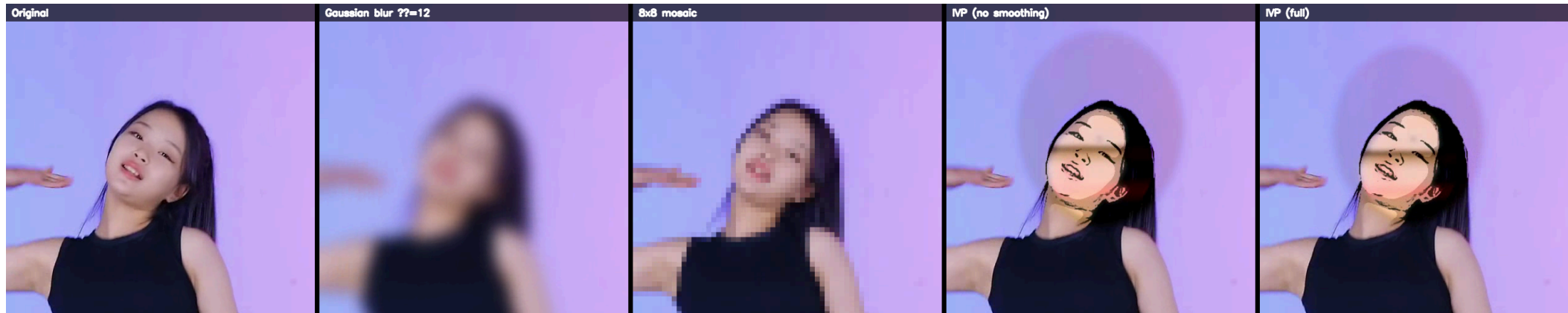
- **One Euro filter** on bbox (cx, cy, w, h) — adaptive low-pass, hold-last-good ≤ 6 frames
- **EMA $\alpha=0.85$** on the 3 Lab K-means cluster centers — posterize bands locked across frames

Ablation (head-region MAD, n=667 frames)

Baseline	MAD
Control	8.29 ± 3.57
IVP OFF	20.17 ± 10.25
IVP ON	17.66 ± 7.48

Excess MAD drops ~30 %; std halved — EMA suppresses outlier "boiling" frames while One Euro keeps the bbox locked even when Haar misses 1–2 frames.

Privacy — FaceNet Re-ID + Detector Survivability

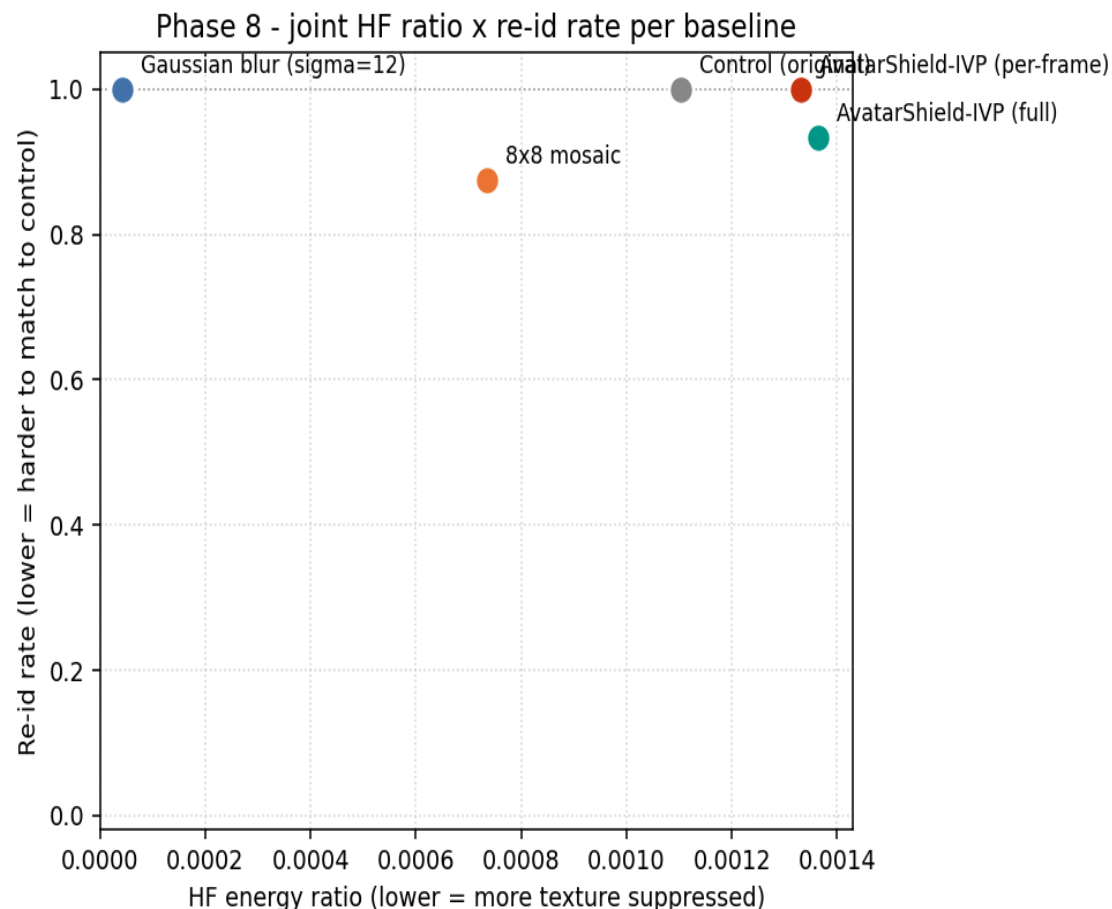


Same frame · control · blur $\sigma=12$ · 8×8 mosaic · IVP no-smooth · IVP full
FaceNet (vggface2), $\cos d > 0.7$, every 15th frame, $n=33$ on `clip.mp4`

Baseline	n (paired)	mean dist	re-id rate
Gaussian blur $\sigma=12$	5	0.596	0.800
8×8 mosaic	17	0.526	0.941
AvatarShield IVP	24	0.673	0.667

- **IVP: lowest re-id (0.667) AND highest detector survivability (24/33)**
- Blur "wins" re-id only because detector dies (5/33) — false win
- IVP keeps head locate-able *and* pushes re-id below blur and mosaic

Joint Trade-Off — FFT HF Ratio × Re-ID Rate



Reading the scatter

- **Blur** — top-left: maximum texture suppression, but detector dies
- **Mosaic** — middle: blocks both axes moderately
- **Control / IVP** — top-right: XDoG line art **is** HF signal by design

Why re-id is the headline

- HF ratio alone rewards blur unfairly
- Re-id with detector-survivability count tells the honest story
- IVP wins the joint trade-off without faking either axis

Note: scatter is Yejin clip (n=20), table on previous slide is clip.mp4 (n=33) — different sources, same conclusion.

Limitations & Non-Claims

What this does NOT claim

- **n=1 source clip** for headline numbers (dancing clip fails when face < 5 % of frame height — Haar limit)
- **Single embedder** — FaceNet vggface2 only; ArcFace deferred
- **No human study** — self-recognition Studies A/B deferred
- **Threat model** — defends *casual* face matching only

Out of scope

- Motivated attacker with reference photo
- Re-identification by people who already know the subject
- Real-time / on-device deployment (2.7 fps CPU)
- Production privacy guarantee (research prototype)

AvatarShield ≠ K-anonymity guarantee. It is **visual-story-preserving anonymization** for child-safe social sharing, defended by transparent classical IVP operators — not by black-box trust.